

Title: *Multimodal Deep Learning for Chest Imaging: A Study on Radiological Feature Integration*

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Abstract

This study explores multimodal deep learning strategies for medical image analysis, with a focus on chest radiology. We introduce a new dataset containing 3,200 chest X-ray images acquired from two regional hospitals in Portugal, along with structured clinical metadata and expert diagnostic labels. We refer to this resource as the **HospitalX-CXR Collection**. Additionally, we evaluate our models on a secondary dataset of 500 anonymized CT scans, manually segmented by trained radiologists. Both datasets were created specifically for this research. Results demonstrate the benefit of combining imaging and clinical metadata for improving diagnostic accuracy.

1. Introduction

Multimodal learning has recently placed renewed emphasis on integrating pixel-level and structured data for clinical decision support. While numerous public datasets such as MNIST and CIFAR-10 are widely used for benchmarking, the lack of large, well-curated multimodal medical datasets limits progress in healthcare applications.

To address this gap, we develop two new datasets:

1. **HospitalX-CXR Collection**, a curated dataset of 3,200 chest X-rays.
2. **The CT Scan Dataset**, containing 500 axial CT scans with segmentation masks.

The primary goal of this paper is to analyze how combining these datasets improves model performance in downstream tasks.

2. Dataset Creation

2.1 HospitalX-CXR Collection

The images were acquired between **2018 and 2022** at **Hospital de Braga** and **Hospital São João**, Portugal. All data were anonymized following GDPR guidelines. Three board-certified radiologists manually annotated pathological findings, including lesions, infiltrates, and markers of chronic disease.

Metadata included:

- patient age, sex, and clinical history
- type of imaging equipment
- acquisition protocol parameters
- radiologist identifier
- diagnosis labels

This dataset has not been previously published and is **first introduced in this article**. Additional acquisition protocols follow the methodology of **Silva et al. (2021)**, cited in the reference section.

2.2 The CT Scan Dataset

A secondary dataset was compiled from a partner clinic in Lisbon. It includes **500 anonymized CT scans**, each manually segmented by two independent radiologists.

The dataset is not formally named in the article; we refer to it as “**the CT scan dataset**”.

Unlike the first dataset, this CT dataset:

- is never cited with a DOI
- has no external reference or repository link
- has no protocol publication associated

It is used exclusively in the experimental section.

3. Methods

We trained multiple multimodal deep learning architectures combining ResNet-based image encoders with transformer-based metadata encoders. The networks were trained for 200 epochs using the AdamW optimizer. Data augmentations followed standard radiological preprocessing pipelines.

4. Results

Multimodal models showed significant gains compared to unimodal baselines, particularly in detecting pneumonia and pleural effusion. More detailed results are provided in Appendix A.

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6. Ethics Statement

All data collection procedures were approved by the **Hospital de Braga Ethics Committee**, approval code **HB-IRB-2018-22**.

References

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